

Farming Under Red Suns

Atmospheric Technosignatures in ExoFarm Scenarios around M-Dwarf Stars

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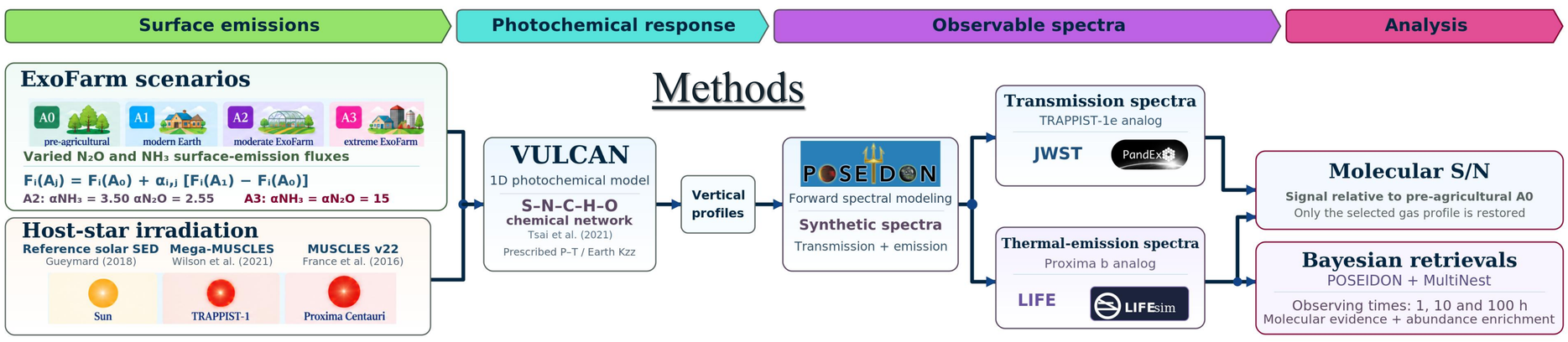
Abstract

Industrial agriculture alters Earth's nitrogen cycle through synthetic fertilizers enabled by the Haber–Bosch process. Agriculture contributes approximately 81% of global NH_3 emissions and 26–30% of N_2O emissions (Wyer et al., 2022; Tian et al., 2024). Similar perturbations on inhabited exoplanets could produce agricultural technosignatures, a possibility explored through ExoFarm (Haqq-Misra et al., 2022), but their atmospheric persistence depends on stellar photochemistry and chemical sinks. We modeled four scenarios, from pre-agricultural to intensified emissions, under solar, TRAPPIST-1, and Proxima Centauri irradiation, evaluating their observability with JWST and LIFE. M-dwarf irradiation enhances NH_3 accumulation in oxidizing, Earth-like atmospheres. While JWST transmission remains challenging, simulated LIFE retrievals for a Proxima b analog provide strong evidence for N_2O at 10 h and moderate evidence for NH_3 at 100 h under extreme ExoFarm conditions. At 100 h, the joint posterior probability of abundances exceeding modern-Earth agricultural levels (A1) reaches 98.67%.

Question

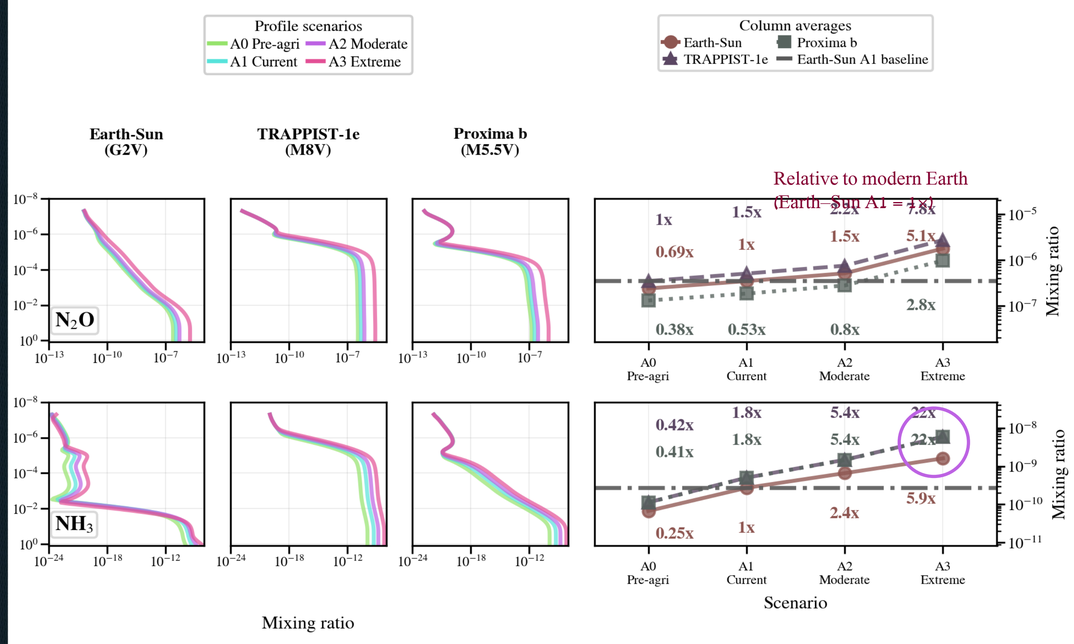
How does stellar irradiation affect the atmospheric accumulation of N_2O and NH_3 from large-scale agriculture, and how well could these gases be detected and retrieved in exoplanet spectra?

Methods



Photochemical Response

Vertical chemical profiles — **Column-averaged mixing ratio**



N_2O — **Broadly tracks agricultural emissions**

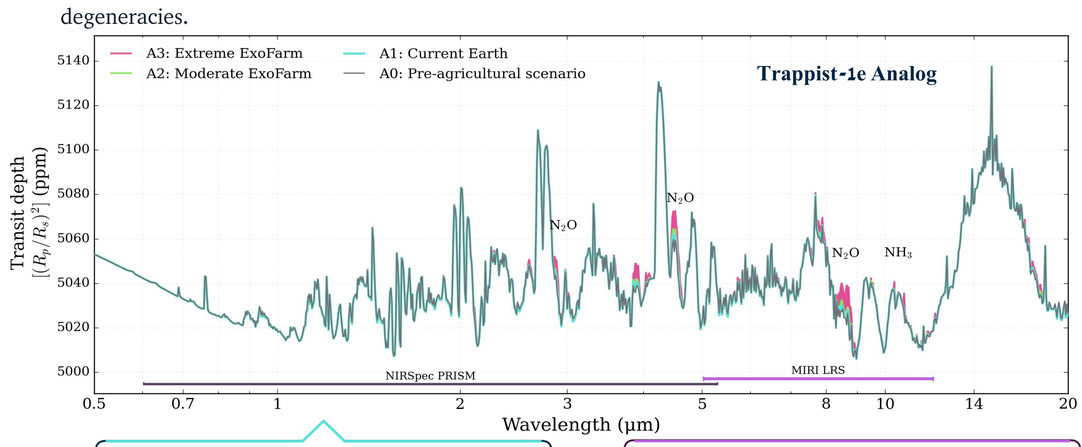
Atmospheric abundances increase approximately with surface forcing, while stellar irradiation shapes their magnitude and vertical distribution.

NH_3 — **Strongly photochemistry-dependent**

M-dwarf irradiation favors NH_3 accumulation and vertical persistence. Under extreme ExoFarm conditions, column abundances reach ~22× the modern-Earth reference.

Spectral Signatures and Detectability

Although A0–A3 spectra largely overlap, their molecular contributions imply very different observing requirements. In the modeled configurations, LIFE achieves greater sensitivity to N_2O than the JWST transmission case, while NH_3 remains substantially more challenging. Molecular S/N assumes known spectral signatures; Bayesian retrievals are needed to assess molecular evidence and atmospheric degeneracies.



JWST · Transmission
TRAPPIST-1e analog · A3

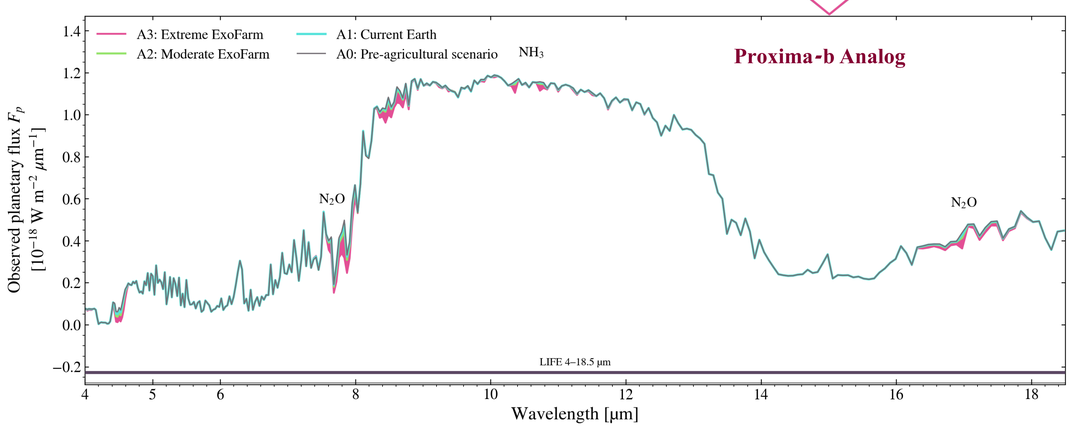
2.9 N_2O S/N

100 NIRSpec PRISM transits
NH_3 remains below S/N = 1 after 100 transits
N_2O S/N = 5: >290 transits (projected)

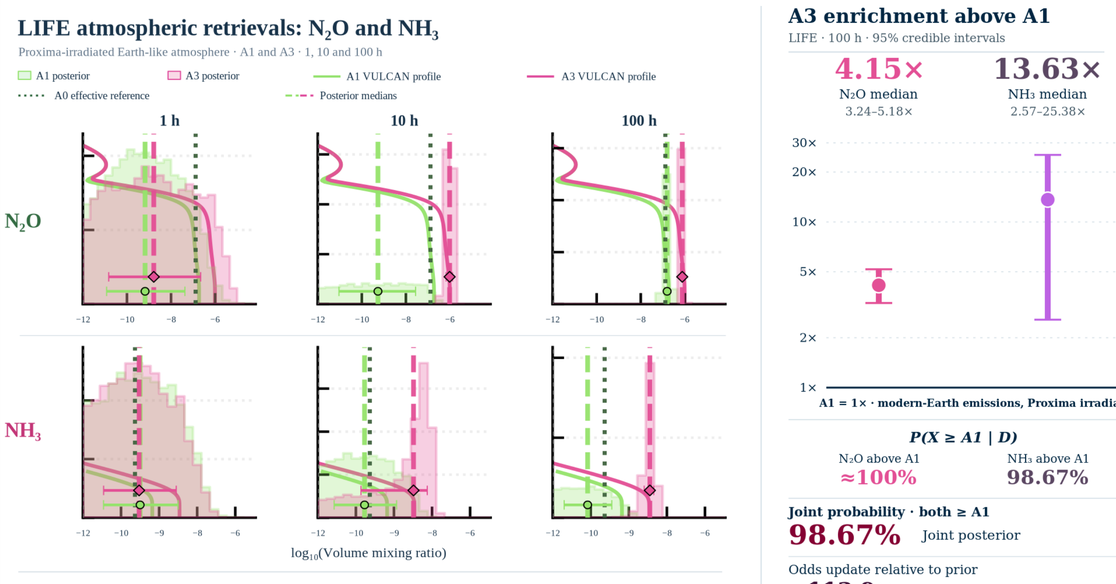
LIFE · Thermal Emission
Proxima b analog · A3 · 10 h

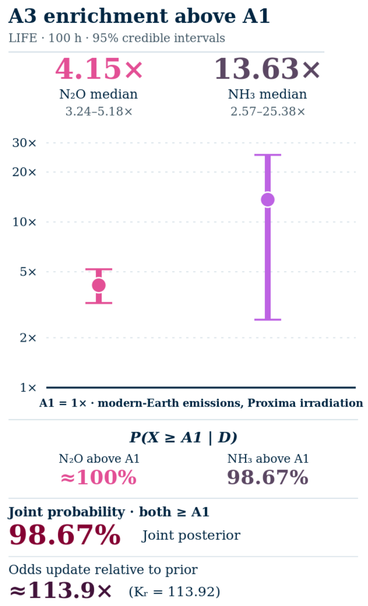
5.97 N_2O S/N | **1.54** NH_3 S/N

Projected for S/N = 5: N_2O ≈ 7 h · NH_3 ≈ 106 h



Bayesian Retrievals and ExoFarm Enrichment





Molecular evidence

ln B · full model vs. molecule-excluded models

		NITROGEN TRACERS		ATMOSPHERIC CONTEXT			
t (h)	Scenario	N_2O	NH_3	H_2O	CO_2	CH_4	O_3
1	A1	-0.549 ± 0.241	-0.627 ± 0.240	12.08 ± 0.24	470.39 ± 0.28	11.10 ± 0.24	-0.18 ± 0.24
1	A3	-0.030 ± 0.235	-0.526 ± 0.234	18.43 ± 0.23	447.76 ± 0.30	5.50 ± 0.23	-0.82 ± 0.23
10	A1	0.416 ± 0.299	-1.865 ± 0.295	224.85 ± 0.30	4285.65 ± 0.37	53.31 ± 0.29	0.10 ± 0.29
10	A3	9.142 ± 0.292	-0.226 ± 0.293	242.08 ± 0.31	4548.02 ± 0.35	27.92 ± 0.29	0.24 ± 0.29
100	A1	9.834 ± 0.341	-0.590 ± 0.344	2252.63 ± 0.35	47460.68 ± 0.43	307.90 ± 0.34	40.13 ± 0.34
100	A3	84.661 ± 0.347	2.749 ± 0.345	1952.51 ± 0.37	39540.78 ± 0.40	274.71 ± 0.34	20.84 ± 0.34

Molecular evidence — In the extreme ExoFarm scenario, LIFE retrievals provide strong evidence for N_2O at 10 h (ln B = 9.14), while NH_3 reaches moderate evidence at 100 h (ln B = 2.75).

Atmospheric enrichment — At 100 h, both gases are inferred above modern-Earth agricultural levels (A1) with 98.67% joint posterior probability. N_2O is more precisely constrained, whereas NH_3 exhibits greater but less certain enrichment.

Take-Home Message



Stellar photochemistry matters

M-dwarf irradiation favors atmospheric accumulation and vertical persistence, particularly for NH_3.



N_2O is the more accessible tracer

LIFE offers promising sensitivity, while JWST transmission remains challenging in our modeled configurations.



NH_3 requires longer observations

Moderate molecular evidence emerges at 100 h under extreme ExoFarm conditions.



Joint enrichment supports intensified nitrogen cycling

At 100 h, N_2O and NH_3 in A3 are inferred above the modern-Earth agricultural benchmark (A1), without uniquely establishing agriculture.

Limitations & Future Work

ExoFarm scenarios prescribe scaled N_2O and NH_3 surface fluxes without a coupled biogeochemical model linking agricultural activity to emissions. Simulations also assume fixed atmospheric thermal structures and vertical mixing, cloud-free spectra, and simplified instrumental noise. Photochemical predictions depend on stellar UV spectra and their spectral discretization (Sneed et al., 2026). Moreover, elevated N_2O may arise naturally under different planetary conditions (Schwieterman et al., 2022), while NH_3 accumulation depends strongly on surface sinks (Huang et al., 2022). Future work should incorporate stellar variability, atmospheric feedbacks, and coupled photochemical–biogeochemical models, ultimately enabling retrievals of surface emission fluxes rather than atmospheric abundances alone.

More Information

Extended Poster · Methods · Full References